

Advanced Data Mining Project

Edge-Ready ECG Classification

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1 Business / Application Understanding

1.1 Application Scenario

Cardiovascular disease remains the leading cause of death worldwide, accounting for an estimated 17.9 million deaths annually, with roughly three-quarters of these occurring in low- and middle-income countries [5]. The electrocardiogram (ECG) is a well-established, non-invasive, and comparatively simple diagnostic tool for detecting cardiac arrhythmias [3], but its clinical use is still largely bottlenecked by manual, beat-by-beat interpretation, which is time-consuming, error-prone, and difficult to scale to long-term or continuous monitoring, particularly for less experienced practitioners [3]. Automated arrhythmia classification directly addresses this bottleneck by allowing large volumes of ECG data to be screened consistently and continuously.

However, the value of an accurate classifier depends heavily on where it can run. The practical impact of ECG classification research is tied to its feasibility on resource-constrained hardware such as Holter monitors, wearable ECG patches, and implantable loop recorders [3], since these are the devices actually capable of providing the continuous monitoring that real-world cardiac care requires. Running inference directly on such a device, rather than relying on a cloud back-end, removes the dependency on a stable network connection, reduces latency, and avoids the power cost of continuously transmitting raw physiological data — all of which matter for a battery-powered wearable intended for long-term use. This motivates this project’s Edge-AI framing: the goal is not simply to build an accurate ECG classifier, but to build one that is deployable on a microcontroller-class device such as the Analog Devices MAX78002.

This project situates itself around three complementary benefits that continuous, on-device ECG monitoring offers over conventional care pathways:

- **Accessibility.** On-device inference does not require a live network connection, which is particularly relevant in rural or low-connectivity settings where cloud-dependent remote monitoring is impractical. Combined with the global and geographic distribution of cardiovascular disease burden [5], this suggests that low-cost, self-contained monitoring devices could extend diagnostic reach to populations currently underserved by conventional, clinic-centered cardiac care.
- **Cost and clinical workload.** Continuous, low-cost home or ambulatory monitoring shifts care from scheduled, staff-intensive inpatient observation toward event-triggered clinical attention, i.e., patients are seen when warning signs actually emerge rather than being monitored in a hospital bed by default. A systematic review of remote patient monitoring (RPM) interventions reports a consistent downward trend in hospital admission and readmission risk, length of stay, and number of outpatient visits associated with RPM [4].
- **Privacy.** Because inference happens on-device, raw ECG data need never leave the device at all — only classification outputs (if anything) would need to be transmitted. This architectural property removes the data-in-transit exposure inherent to cloud-based analysis and aligns naturally with data-minimization principles common in health-data handling, without requiring any claim of formal regulatory compliance, which is out of scope for this project.

Potential users of such a system include patients requiring long-term monitoring following a cardiac event, clinics or regions lacking on-site cardiologist coverage, and clinical workflows where an automated system pre-screens ECG recordings to flag likely arrhythmic events for a clinician’s review.

This system is not intended as a certified diagnostic device and no claim of clinical validation is made; its role is to flag, not to diagnose.

1.2 Data Mining Objective

The machine-learning problem addressed in this project is formally defined as follows.

Input: A fixed-length discrete signal segment containing a single localized heartbeat from a single ECG channel (centered around the R-peak).

Output: one class label from the set $\{N, S, V, F\}$, corresponding to a four-class simplification of the AAMI EC57 superclass scheme, obtained by excluding the rare paced (P) and unclassifiable (Q) beat categories, as specified in the project brief.

Type of task: Supervised multiclass class-imbalanced time-series classification.

1.3 Project Objectives

The technical objectives of this project are:

- Accurately classify individual ECG beats into N, S, V, and F categories.
- Ensure model generalization to unseen patients by enforcing a strict record-wise train/test split.
- Maintain classification accuracy under real-world noise conditions (baseline wander, muscle artifacts).
- Develop an ultra-compact neural network fitting within the 2MB memory constraint of the MAX78002.
- Ensure structural compatibility with the Analog Devices MAX78002 CNN accelerator (using supported 1D convolutions, pooling, and INT8 quantization).

1.4 Success Criteria

Success is defined across five dimensions, each with a measurable target rather than a general aspiration toward "high accuracy."

Predictive performance. Overall accuracy is discarded in favor of macro-averaged F1-score, Sensitivity (Se), and Positive Predictive Value (+P) to account for severe class imbalance [3].

Minority-class performance. The model must achieve realistic sensitivity and positive predictive value targets for the rare S and F classes under a strict inter-patient evaluation protocol.

Noise robustness. Defined as a bounded degradation in macro-F1 across clean, moderate (6 dB), and strong (0 dB) noise conditions. Because distinct noise sources damage the ECG uniquely (e.g., Gaussian noise erases waveform features, electrode-motion corrupts P/T waves, and muscle artifacts distort structure while preserving the R-peak [1]), robustness is evaluated independently against specific noise profiles from the MIT-BIH Noise Stress Test Database rather than a single blended metric.

Model complexity. The model must strictly adhere to the 2 MB memory ceiling. Because embedded ECG models in the literature typically range from 93 KB (aggressively pruned) up to several megabytes [3], a sub-1 MB quantized footprint serves as our realistic target band beyond the hard project constraint.

MAX78002 compatibility. Success is strictly binary. The model must only use operations supported by the MAX78002 CNN accelerator (1D convolution, pooling, ReLU, compact FC layers) and pass software-level INT8 quantization and synthesis with bounded accuracy degradation relative to the FP32 baseline.

1.5 Constraints

Data-level constraints. The cohort is small (47 subjects) and severely imbalanced, with N beats comprising 82.78% of annotations versus just 2.54% for SVEB and 0.73% for F beats [3]. Furthermore, the secondary ECG lead identity varies across subjects (e.g., V1, V2, V4, V5) due to differing electrode placements [3], preventing a fully homogeneous two-lead input and limiting broad generalization claims.

System-level constraints. The target platform requires MAX78002-compatible INT8 computation and imposes a hard 2 MB limit. Without a physical development board, all embedded-feasibility metrics (e.g., inference latency, power consumption) are strictly limited to software-level estimates rather than measured on-chip outcomes.

Methodological uncertainty as a constraint. Standard ECG preprocessing cannot be assumed beneficial by default. Prior comparative research shows band-pass filtering yields no measurable classification improvement, and min-max normalization can actually be mildly detrimental [2]. Consequently, conventional preprocessing steps must be empirically ablated rather than blindly applied.

1.6 Risks and Challenges

Risk	Why it matters	Proposed mitigation
Data leakage via intra-patient splitting	Only 30.3% of reviewed studies use correct inter-patient splitting; improper intra-patient splitting artificially inflates accuracy (e.g., from 88.34% to 99.48%) [3].	Enforce the strict record-level DS1/DS2 split; verify no record ID appears in both training and test sets.
Minority-class metrics collapsing	Under honest inter-patient evaluation, SVEB sensitivity often collapses to 75–77% even in well-regarded models [3].	Report per-class Sensitivity/PPV continuously; prioritize minority-class recall on the DS2 test set as a headline metric.
Unmeasurable quantization loss	FPGA deployments demonstrate accuracy drops (e.g., 90.8% to 86.4%) purely due to fixed-point quantization effects [3].	Compare FP32 and simulated-INT8 results side by side; explicitly label all on-chip figures as unvalidated software estimates.
Noise robustness overfitting	Different noise types damage waveforms uniquely (e.g., Gaussian erases features, while muscle artifact preserves the R-wave) [1].	Evaluate against multiple distinct noise types from the MIT-BIH Noise Stress Test Database rather than a single blended metric.
Ineffective preprocessing	Conventional filtering and min-max normalization often fail to improve downstream classification or actively degrade it [2].	Ablate each preprocessing step (with/without) rather than applying standard ECG preprocessing by default.

2 Data Understanding

2.1 Dataset Description

- **Database:** MIT-BIH Arrhythmia Database.
- **Records Used:** 44 fully annotated, approximately 30-minute recordings (divided equally into a 22-record training set and a 22-record test set).
- **Subjects:** 44 distinct patients.
- **Channels:** 2. The upper channel is used for extraction. Batch extraction confirms this is Lead II (MLII) for 43 records, and Lead V5 for 1 record.
- **Sampling frequency:** 360 Hz.
- **Annotations:** Expert-verified heartbeat peak locations and classifications.

2.2 Target Classes

The original MIT-BIH dataset utilizes 23 distinct symbols to annotate heartbeats, rhythm changes, and signal quality. To standardize the classification task for the required N, S, V, and F target classes, these raw symbols were extracted and mapped according to the standard Association for the Advancement of Medical Instrumentation (AAMI) EC57 guidelines.

Empirical extraction of all annotation files in the dataset yielded the exact raw symbol distributions, mapped to the final classes as follows:

Final class	Original annotations included	Number of beats extracted
N (Normal)	N (75052), L (8075), R (7259), j (229), e (16)	90,631
S (Supraventricular)	A (2546), a (150), J (83), S (2)	2,781
V (Ventricular)	V (7130), E (106)	7,236
F (Fusion)	F (803)	803
Excluded	/, +, f, ~, !, ", x, —, Q, [,]	11,260

Table 1: Mapping of empirical raw MIT-BIH symbols to AAMI target classes.

Non-beat annotations (such as ‘+’ for rhythm changes and ‘~’ for signal quality changes) and paced/unclassifiable beats (such as ‘/’ and ‘f’) were explicitly excluded from the dataset prior to model input generation because they do not belong to the four required target classes.

2.3 Exploratory Data Analysis

Batch processing of the 44 records into the designated DS1 and DS2 splits reveals extreme class imbalance and cross-split variance. Out of 100,718 total valid heartbeats extracted, the Normal (N) class heavily dominates, while Fusion (F) beats account for less than 1%.

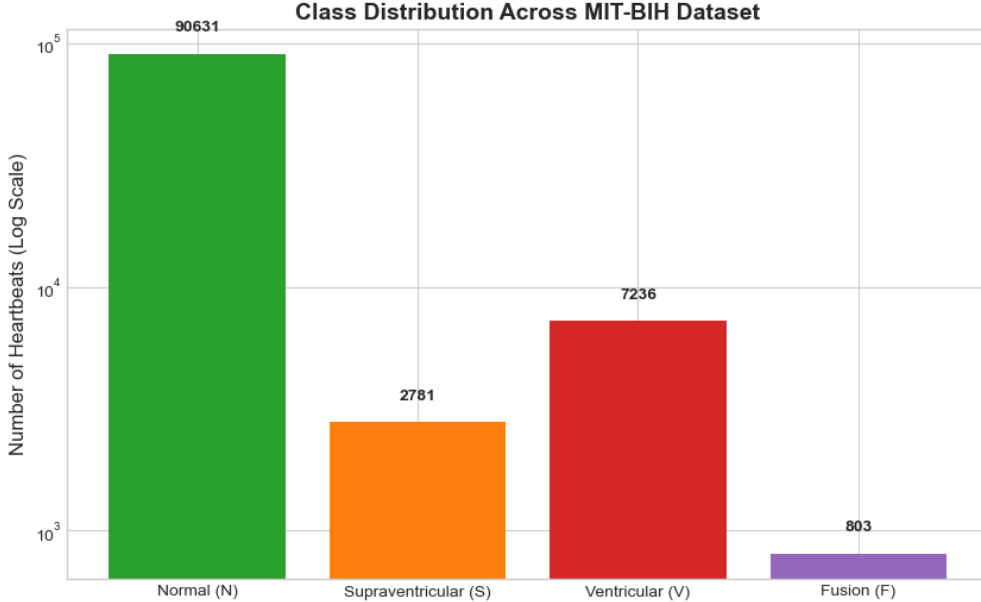


Figure 1: Logarithmic class distribution across the MIT-BIH dataset highlighting severe pathological imbalance.

Crucially, analyzing the dataset across the standard DS1 (train/val) and DS2 (test) splits shows that the minority class distributions are not uniform across patient cohorts:

- **N (Normal):** 89.91% in DS1 (45,866) vs. 89.04% in DS2 (44,259).

- **S (Supraventricular):** 1.85% in DS1 (944) vs. 3.70% in DS2 (1,837).
- **V (Ventricular):** 7.43% in DS1 (3,788) vs. 6.48% in DS2 (3,221).
- **F (Fusion):** 0.81% in DS1 (415) vs. 0.78% in DS2 (388).

The prevalence of class S beats exactly doubles from the training set to the test set, highlighting a severe distribution shift. This EDA finding explicitly dictates the need for loss-weighting mechanisms and proves that standard random-split accuracy would yield fundamentally misleading evaluation metrics.

To evaluate the morphological separability of the classes, representative heartbeats were isolated and normalized across the four categories, as shown in Figure 2. The extracted segments show clear structural differences: Ventricular ectopic beats (V) present a visibly widened and inverted QRS complex, while Supraventricular beats (S) display altered P-wave timing, confirming that morphological 1D convolutional feature extraction is well-suited for this classification task.

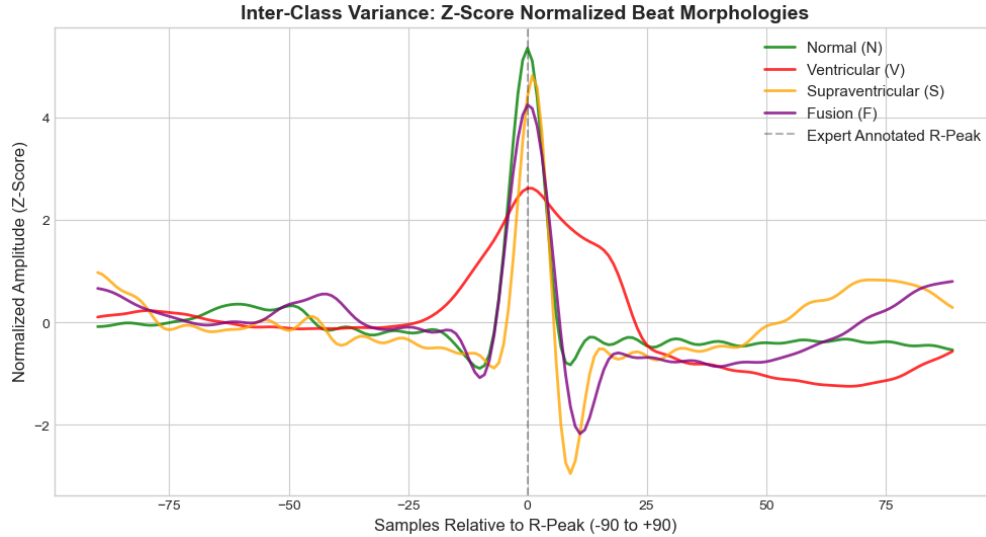


Figure 2: Inter-class morphological variance demonstrating an example heartbeat from each of the four target AAMI classes.

2.4 Data Quality

Batch inspection of the 44 utilized records confirms high overall data quality, validating our preprocessing assumptions:

- **Missing values:** No NaN (missing) values were detected in any of the primary signal arrays.
- **Amplitude outliers:** No extreme amplitude spikes exceeding basic physiological recording thresholds (-1000 to +3000 ADC units) were detected, indicating that complex algorithmic signal clipping is unnecessary.
- **Differences between ECG channels:** 43 records utilize MLII as the upper channel, while 1 record utilizes V5. This finding mandates dynamic channel selection during preprocessing to prevent crashing, while maintaining the validity of a single-channel neural network architecture.

Corrupted Segments and Noisy Regions: While general data completeness is high, individual ambulatory recordings exhibit severe localized artifacts caused by physical sensor movement and muscle activity. Figure 3 compares a stable baseline recording against a severely degraded segment from Record 108. The presence of baseline drift and high-frequency electromyographic (EMG) noise in real recordings reinforces why filtering is strictly required and justifies our planned noise-robustness experiments on held-out data.

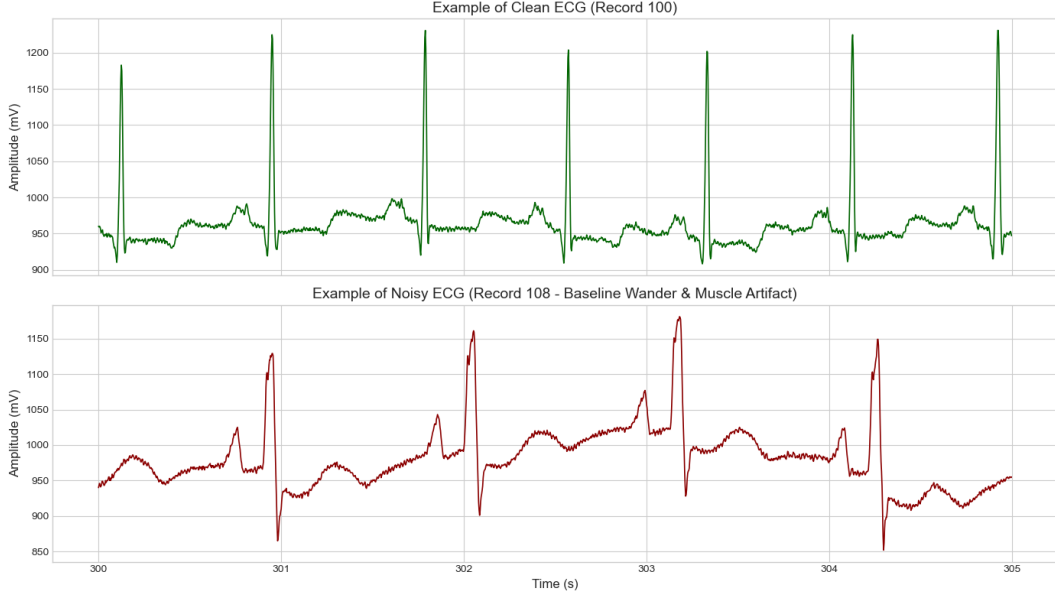


Figure 3: Comparison of a clean ECG recording (Record 100) against a noisy segment corrupted by baseline drift and muscle artifacts (Record 108).

2.5 Leakage Analysis

Because ECG morphology is highly patient-specific, randomly dividing a single patient’s beats into both training and testing sets (intra-patient splitting) causes the model to overfit on biological identity rather than learning general diagnostic features.

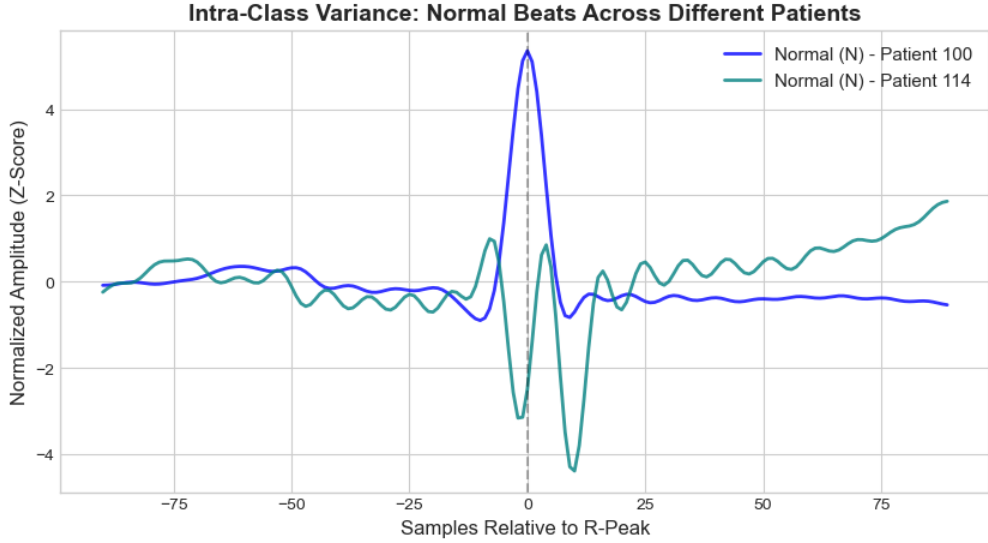


Figure 4: Intra-class variance demonstrating morphological differences between recordings for the same Normal (N) target class.

As shown in Figure 4, a Normal (N) beat from Patient 100 differs fundamentally in peak amplitude and repolarization shape from a Normal (N) beat from Patient 114. This intra-class variance confirms that random splitting across beats would lead to catastrophic data leakage. A strict inter-patient splitting protocol was applied. Empirical batch verification of the record IDs confirms exactly **0 patient overlap** between the DS1 and DS2 sets. This absolute separation guarantees that the downstream evaluation will honestly reflect the model’s out-of-distribution generalization.

3 Data Preparation

3.1 Record Selection

To guarantee rigorous generalization assessment, the dataset was strictly partitioned at the patient (record) level. The 22 records from the DS1 development set were further subdivided into training and validation sets to allow for proper hyperparameter tuning without touching the final DS2 test set:

- **Training (17 records):** 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 209, 215, 220, 223, 230.
- **Validation (5 records):** 201, 203, 205, 207, 208.
- **Test/DS2 (22 records):** 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234.

Figure 5 visualizes the distribution of valid beats across individual patient records. The beat volume varies significantly between subjects (from under 1,500 to nearly 3,000 beats). This non-uniformity confirms that pooling data would allow high-volume records to dominate gradient updates, providing further technical justification for our record-independent partition.

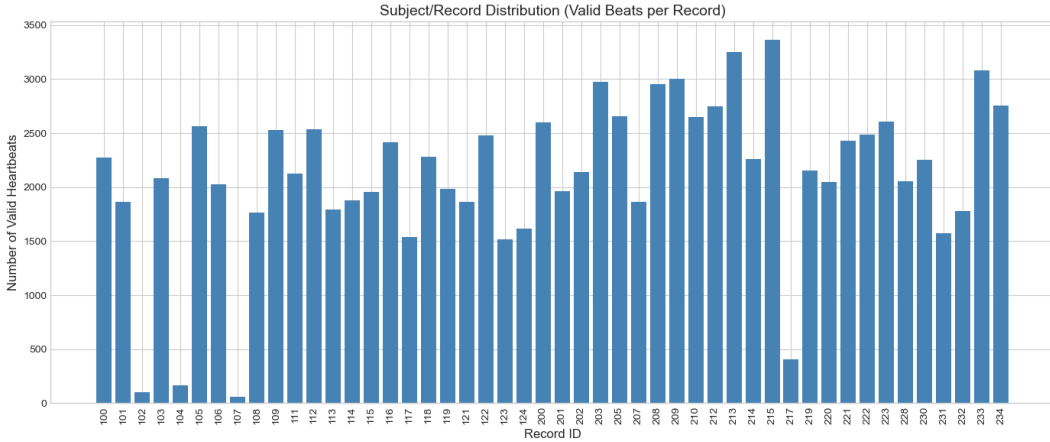


Figure 5: Subject/record distribution showing the non-uniform count of valid heartbeats extracted per patient record.

3.2 ECG Channel Selection

The upper channel is designated as the primary signal source. Batch inspection proved this is Lead II (MLII) for the vast majority of records, with the exception of one record utilizing Lead V5. The extraction logic dynamically defaults to the available upper channel array index. This maintains a uniform single-channel input pipeline suitable for Edge-AI deployment without arbitrarily discarding patient data.

3.3 Beat Extraction & The Deployment Gap

Individual heartbeats are extracted using a fixed-width temporal window:

- **Samples before peak:** 90
- **Samples after peak:** 90
- **Final input length:** 180 samples (0.5 seconds at 360 Hz)

At 360 Hz, 180 samples equates to exactly 0.5 seconds. This width captures the complete P-QRS-T complex while remaining narrow enough to avoid overlapping with adjacent beats during tachycardia.

Dual-Path Extraction Methodology: A known limitation in academic ECG classification is the reliance on expert annotations during inference. A real Edge-AI wearable relies on a continuous algorithmic R-peak detector prior to classification, introducing temporal jitter (off-center peaks), missed beats (false negatives), and noise-induced ghost beats (false positives).

To rigorously quantify this deployment gap, an empirical batch evaluation was conducted across the entire dataset utilizing the `NeuroKit2` algorithmic peak detector. Compared against the 109,494 MIT-BIH ground-truth annotations, the algorithmic detector produced 1,782 missed beats (1.63%), generated 2,441 ghost beats (2.22%), and exhibited an average temporal jitter of 1.23 samples (3.42 ms) on successfully matched beats.

Because evaluating the neural network solely on algorithmically detected peaks confounds morphological classification errors with upstream detection errors, the DS2 test set will be processed using two distinct pipelines:

1. **Theoretical Baseline (Expert Extraction):** Extraction utilizing the MIT-BIH expert annotations. This strictly isolates the neural network’s intrinsic classification performance from detection errors.
2. **Edge-AI Simulation (Algorithmic Extraction):** Extraction utilizing the automated `NeuroKit2` R-peak detector. This evaluates true end-to-end performance, measuring exactly how the previously quantified temporal jitter and missed beats degrade final classification accuracy in a simulated deployment environment.

3.4 Signal Preprocessing

Because the Edge-AI Simulation path relies on automated R-peak detection, robust signal preprocessing is strictly required. Preprocessing operations are performed sequentially in the following explicit order:

1. **Lead Selection and Loading:** The 1D continuous signal array is extracted from the upper channel (MLII or V5). *Necessity:* Establishes a standardized single-lead input for the downstream neural architecture.
2. **FFT Bandpass Filtering (0.5–40.0 Hz):** A fast Fourier transform (FFT) bandpass filter isolates frequencies between 0.5 Hz and 40.0 Hz. *Necessity:* The 0.5 Hz highpass threshold eliminates respiratory baseline wander. The 40.0 Hz lowpass threshold eliminates high-frequency electromyographic (EMG) noise and mains hum without distorting the clinical QRS complex.

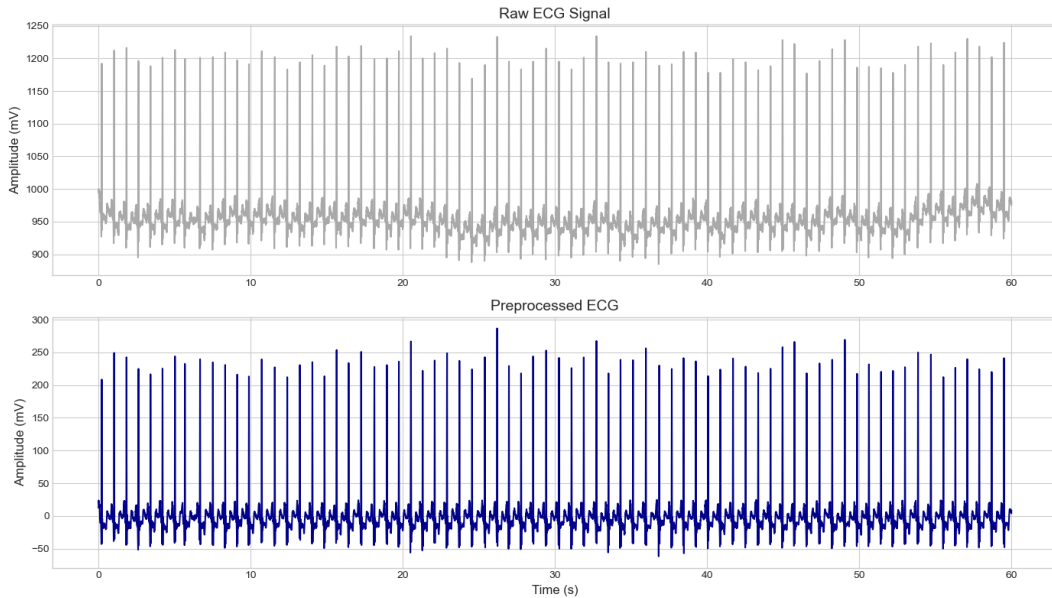


Figure 6: Continuous ECG signal before and after applying the 0.5–40.0 Hz FFT bandpass filter, highlighting baseline stabilization and high-frequency smoothing.

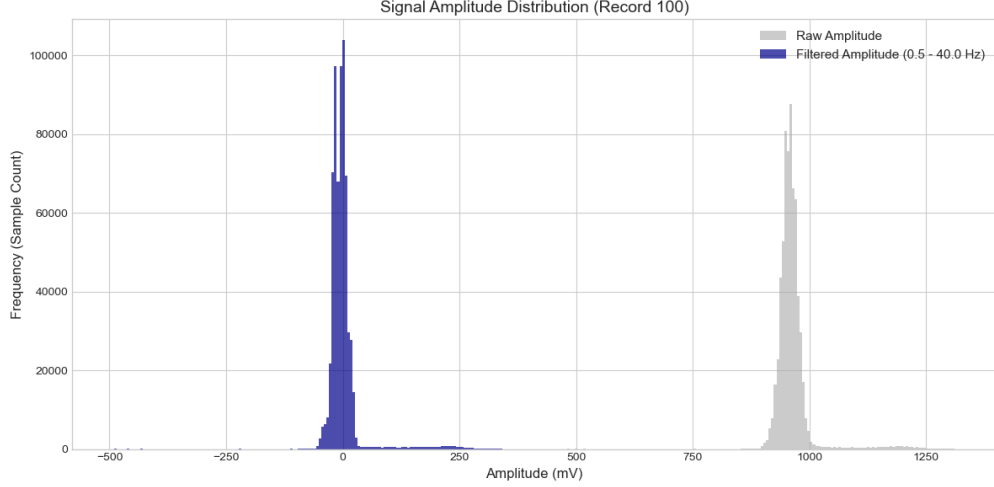


Figure 7: Signal amplitude distribution comparing the raw ECG voltage against the stabilized, zero-centered distribution post-filtering.

3. **Beat Window Segmentation:** The continuous filtered signal is sliced into fixed 180-sample segments centered on the detected peak index. *Necessity:* Embedded neural networks require fixed-dimension tensors.
4. **Per-Beat Z-Score Normalization:** Each extracted 180-sample beat x is normalized independently according to:

$$x_{norm} = \frac{x - \mu}{\sigma}$$

Necessity: As seen in Figure 7, bandpass filtering centers the global signal, but individual beat voltages still vary across subjects due to skin impedance. Per-beat normalization ensures uniform scale and prevents saturation during INT8 quantization on the MAX78002 accelerator.

3.5 Label Preparation

Raw string annotations are parsed and deterministically mapped to the four clinical AAMI superclasses (N, S, V, F) as defined in Section 2.2. All unmappable symbols (rhythm/quality markers and paced beats) are silently bypassed during the extraction loop, generating a clean target array aligned with the extracted signal windows.

3.6 Class Imbalance

The training dataset suffers from extreme pathological imbalance. Normal (N) beats total 35,994, while Fusion (F) beats comprise merely 28 samples in the training split. Standard cross-entropy loss would cause the network to collapse into a majority-class classifier, yielding high overall accuracy by defaulting to the N class while failing entirely to detect rare arrhythmias.

To counteract this, the training pipeline utilizes Focal Loss. Focal Loss addresses severe foreground-background imbalance by dynamically down-weighting the penalty for easily classified examples, forcing the model to focus gradient updates on hard, misclassified minority beats. The mathematical formulation is:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Where p_t is the model’s predicted probability for the ground-truth class, γ is the focusing parameter (commonly set to 2.0), and α_t is a static class weight inversely proportional to class frequency. As the network learns to easily identify N beats, Focal Loss drives the gradient penalty for those easy examples near zero, reserving the model’s limited parameter updates for the difficult S, V, and F beats.

Alternative balancing techniques were rejected. Data-level oversampling methods like SMOTE generate synthetic waveforms through interpolation, which produces biologically implausible morphologies

in physiological time-series data. Undersampling the majority class was also rejected, as discarding tens of thousands of real Normal beats destroys the natural intra-class morphological variance required for the model to generalize across different patients.

3.7 Data Augmentation

To prevent overfitting on the limited 17-patient training cohort and to proactively bridge the deployment gap identified in Section 3.3, data augmentation is applied dynamically to the training split on the fly.

To simulate the physical realities of wearable Edge-AI deployment, the following augmentations are applied:

- **Temporal shift (Algorithmic simulation):** The signal window is randomly translated relative to the true R-peak by an integer offset sampled uniformly from a $[-5, 5]$ sample range. Empirical evaluation proved the automated algorithmic detector introduces an average temporal jitter of 1.23 samples. This augmentation forces the convolutional layers to learn translation invariance, proactively preparing the classifier for the temporal misalignment it will face in the simulated deployment pipeline.
- **Amplitude scaling:** The entire 180-sample vector is multiplied by a scalar value randomly drawn from a uniform distribution between $[0.9, 1.1]$. This explicitly simulates fluctuating sensor contact quality and variations in skin-electrode impedance that dynamically alter signal voltage in ambulatory settings.

3.8 Final Dataset

Following extraction, mapping, and strict inter-patient subdivision, the final dataset contains exactly 100,693 valid heartbeats. The internal validation split (5 records) guarantees the model can be tuned without leaking information from the final DS2 test set.

Split	Records	N	S	V	F	Total Beats
Training	17	35,994	702	1,873	28	38,597
Validation	5	9,862	242	1,915	386	12,405
Test (DS2)	22	44,246	1,837	3,220	388	49,691
Total	44	90,102	2,781	7,008	802	100,693

Table 2: Final class distribution across strictly separated inter-patient data splits (utilizing expert annotations).

3.9 Reproducibility

To ensure all preprocessing steps, ablation studies, and evaluation metrics are fully reproducible, the project adheres to the following constraints:

- **Language/Framework:** Python 3.10+, PyTorch, NeuroKit2
- **Random Seed:** 42 (Strictly enforced across NumPy, PyTorch, and Python random modules)
- **Sampling Frequency:** 360 Hz (Native PhysioNet frequency; no resampling applied)
- **Tensor Shape:** 1×180 (Single channel, 180 temporal steps)

References

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